MMT39 — Final Year-1 Strategy, Graded Round

Company 3 · markets A and B · written 6 September 2026 · supersedes *YEAR 1 - OPTIMAL STRATEGY.txt* and *06-graded-year1-plan-2.1*. Built on the Year-1 test results verified in this repository and the Year-2 test readings carried by Plan 2.1, after those readings passed four independent consistency checks.

The plan in one line. Install **10 S + 20 H lines on three shifts** (€51,000,000, two slots held back), price **S at €18.00/18.50 and H at €8.00 in A and €8.75 in B**, fund the fleet properly, borrow €27,000,000 against the February cash trough, and spend €481,800 on research. Central projection: **net income €164,469,390, ROE 71.4%**, and **€686,475,039 of accumulated profit over four years**.

1 · Grading the two input plans

Both were run through one engine: the Year-1-calibrated cost model, the demand curves fitted to the test-round readings, and identical financing rules. Scores are out of 10.

Area	Plan A my first sheet	Plan B colleague 2.1	What decided it
Evidence discipline	9	7	A verified every figure but had no Year-2 data; B has Year-2 but quotes Y1 S demand as 9,199,028 where the result sheet says 8,419,028, and costs the plan on the uncalibrated model
Pricing	5	3	A's H at €7.33 was directionally right but set without the Y2 evidence; B's H at €5.25 in market A is the single largest error in either plan
Capacity and mix	4	8	A's 22 lines are too small against measured demand; B's 32 lines are right-sized but spend every slot in year one
Cash and financing	8	8	A avoids debt but leaves lines unbought; B sizes the loan to a real monthly trough — the better method
Logistics	7	9	B is the best of the three: its trunk rule is validated against four stockout flags and its last-mile floor is derived from the Y2 failure
Human resources	9	5	A correctly refuses the training department and the wage rises; B buys the €722,000 department against €306,000 of measured turnover cost
Information	8	5	B declines surveys 19, 15, 11 and 4 — the four that size Year 2 and that the rubric asks for by name
Robustness	7	7	Neither states what happens if the demand reading is wrong
Weighted grade	**6.6**	**6.5**	**Different strengths, and the final plan takes the better half of each**

Like-for-like: the same three plans, one engine, six demand scenarios

Plan	worst case	central	best case	mean
Plan A — my first sheet (6S+16H, no Y2 data)	84,013,911	84,335,515	84,717,492	84,409,240
Plan B — colleague 2.1 (8S+24H)	103,937,942	103,937,942	103,937,942	103,937,942
Plan C — FINAL (10S+20H)	**132,683,215**	**164,469,390**	**172,897,815**	**156,933,809**

Plan A and Plan B show no spread because both price low enough that demand exceeds production in every scenario — they sell their whole plant whatever happens, which is precisely the symptom of underpricing.

Where the €60,531,448 gap to Plan B comes from

Change, applied one at a time to Plan B	Net income	Delta
Plan B as written	103,937,942	—
Training department off	104,659,942	+722,000
Full survey programme (19, 15, 11, 4 added)	104,344,942	−315,000
H price in market A, 5.25 → 8.00	129,758,803	+25,413,861
H price in market B, 7.75 → 8.75	136,189,838	+6,431,035
S price 14.50 → 18.00	151,745,957	+15,556,119
Mix 8S+24H → 10S+20H	163,296,126	+11,550,169
S price in B 18.00 → 18.50 (final grid)	**164,469,390**	**+1,173,264**

Three price lines carry 78% of the improvement. Every other decision in both sheets, added together, is worth less than the H price in market A alone.

2 · The evidence, and how far it can be trusted

Plan 2.1's Year-2 readings could not be verified directly — none of the files it cites for them exist in this repository. They were therefore tested against rules that *were* verified from the Year-1 result sheets. All four checks passed, so the readings are treated as sound.

Check	Verified rule	Result
Trunk stockout flags, 4 of 4	vehicle capacity = 500 × load kg	Y1 A no / Y1 B yes / Y2 A yes / Y2 B no — all four predicted correctly
Y2 depreciation 7,056,000	12% of lines and SMED only, land excluded	12% × (52,800,000 + 6,000,000) = 7,056,000 exactly
Y2 plant reconstruction	line prices 2.1M / 1.5M	52,800,000 = 8 S + 24 H exactly
Y2 unit accounting	SMED utilisation 92.9%	H produced 40,132,800 less 12,253,570 unsold leaves S sold inside S capacity

The four readings the plan is built on

Reading	Value	What it fixes
H, market A, price 5.25	68.6% share	the A demand anchor
H, market B, price 7.75	55.1% share	the B demand anchor, and the only high-price H observation
S, market A, price 14.50, 12 insertions	44.8% share	the S demand anchor
S, market B, price 14.50, 3 insertions	~2% share	the cost of starving a market of media

The correction that matters most. Reading the Year-1 results alone, market B outsold market A on half the advertising, and I concluded that spending above roughly 99% net rating buys nothing. The Year-2 readings show the opposite at the bottom end: market B's sunscreen share collapsed from 48.6% to about 2% when its media was cut to three insertions. The response is a saturation curve — flat above roughly seven insertions, cliff-edged below four. **Never starve a market. Match A and B.**

The implied elasticities, and why the plan does not lean on them

- **H share elasticity –0.563** from the two same-year price points. Below 1 in absolute terms, which means revenue rises with price. The confound is that B is the richer market, so the true figure is more negative; the plan is tested from –0.563 to –1.60.
- **S elasticity –1.52** from Y1 to Y2, but that window also contains the media collapse in B, so it overstates the price effect. Tested to –2.50.

The argument that does not need an elasticity at all. At €5.25 we know market A's H share is 68.6%. Ask instead how far that share must fall when the price goes to €8.00 before €5.25 would have been the better choice:

A's H share at the higher price	68.6%	55%	45%	35%	25%
Net income versus €105,219,140 at 5.25	+53.8M	+53.8M	+44.5M	+31.0M	+17.6M

Market A's share would have to fall below **25%** — worse than any outcome observed in either test year, in either market — before the low price wins. That is why the price rise is the recommendation even though the elasticity is uncertain.

3 · The plan, field by field, in interface order

1 Commercial distribution

Market	A	B	E
Sales managers	6	6	0

Own every zone. At €333M of retail sales the wholesaler alternative costs 8.46% = €28,183,330 against roughly €8,000,000 for our own network. Market E stays out: it produced 112,814 units in test Y2 for about €1,600,000 of entry cost. This field resets to zero every year.

2 Logistics

Field	A	B	E	Why
Platforms	12	12	form minimum	the tested configuration
Trunk vehicles	2	2	form minimum	capacity 24,000,000 units each against 17,152,543 in A and 15,774,297 in B — 40% and 52% headroom
Trunk load capacity (kg)	24,000	24,000	form minimum	cheapest per unit at full load
Last-mile vehicles	30	20	form minimum	Y2 evidence floor: 10 vehicles at 10,000 kg stocked out on 12.2M units in B
Last-mile load capacity (kg)	10,000	10,000	form minimum	§12 maximum

Trunk cost €211,200, last mile €1,800,000, platform and warehouse cost about €1,100,000. Zero is not a legal entry anywhere in this form; market E is exited by leaving its price blank, not by zeroing logistics.

3 Human resources

Field	Value	Why
Level I / II / III / IV	126,000 / 84,000 / 60,000 / 35,000	test values; 4 office leavers cost €96,000, a 5% rise would cost €147,000
Operator / Specialist	13,500 / 24,000	output hit exactly 81.25% of nameplate at these wages, so there was no productivity loss to buy off
Recruitment and training department	No	€722,000 against €306,000 (Y1) and €276,000 (Y2) of measured turnover cost. It cannot pay
Reps per zone, A	S 4 · T 4	the configuration that produced the A demand readings
Reps per zone, B	S 5 · T 3	B supermarkets are 37% of volume
Rep salary, both markets	27,000 fixed · 0.60% variable	held at the tested level; commission is paid on channels S and T only

4 Market research — €481,800

Surveys 1, 2, 3, 4, 8, 11, 15, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28.

Added beyond Plan 2.1	Cost	Why it is worth it
19 — demand by company	110,000	sizes the Year-2 line and SMED decision, and is the only view of rival demand

15 — shelf space achieved	72,000	we have asked for shelf space three times and never once seen what we got
4 — sales by company, market and channel	72,000	fixes the channel weights, which are still Year-1 rationed figures
11 — insertions by company	36,000	share of voice, which the rubric names and which the Y2 collapse proved matters
24, 25, 26 — industry cost, salary, balance sheets	25,000	the "compared to other companies" evidence the grading demands

€481,800 is 0.16% of net revenue. 65% of the course grade is analysis, not result.

5 Prices and 8 Trade

Product	Market	MSRP	Margin	Shelf %	POP per channel
S	A — all three channels	18.00	1.90	18	3,000
S	B — all three channels	18.50	1.90	18	3,000
H	A — all three channels	8.00	0.95	20	5,000
H	B — all three channels	8.75	0.95	20	5,000
Market E: every field blank. A price with no shelf space burned €184,000 of E advertising for zero sales in test Y1.					

Flat within each market, deliberately. The demand readings were produced by flat per-market pricing, flat margins, 18% and 20% shelf space and this POP level. Changing any of them invalidates the forecast they support. Channel differentiation is a Year-2 experiment, not a Year-1 guess.

Floors cleared with room: S standard cost 5.5174 plus margin gives a dumping floor of 7.42 against a price of 18.00; H 2.2928 plus margin gives 3.24 against 8.00. Prices sit 3% above the highest H price ever observed (7.75 in B) and 24% above the highest S price observed (14.50) — the S step is justified because at 14.50 demand exceeds any plant the cash allows, so we would be selling capacity, not demand.

6 Promotions — none

Both products are supply-capped at these prices: S demand 6,723,103 against 5,850,000 of output. §48 rejects a promotion that supply cannot cover, so a promotion could only displace a full-price unit with a discounted one. Revisit in Year 2, when SMED lifts output 14%.

7 Advertising

Product	Campaign	Press per market	TV per market	Cost per market	Net rating
S	18	4	8	204,000	99.99%
H	19	3	5	135,000	99.86%
Total media, both markets				**678,000**	**POP 48,000**

Identical in A and B — this is the correction the Year-2 collapse forces. Display, radio and social stay at zero; television is the cheapest reach at €300 per point and social the dearest at €500. Campaign 18 puts sunscreen on the Technological segment and 19 puts moisturiser on Natural, splitting the two 40% segments instead of competing for one; in test Y1 three of five teams chose no campaign at all and took the neutral positioning §17 assigns by default.

9 Production

Field	S	H
Lines to install	10	20
Total lines / lines to be ACTIVATED	10 / 10	20 / 20
Planned output at 81.25%	5,850,000	29,250,000
Raw material to purchase	6,265,350	29,835,000
Opening raw material / finished goods	0 / 0	0 / 0

Shifts	3	labour is charged per line-shift and sits inside standard cost, so cost per unit is identical at 1, 2 or 3 shifts; the only cost of the third shift is 2 points of depreciation on the lines
Preventive maintenance	Yes	81.25% observed against 78.75% without it — 2.5 points of nameplate for €500,000
Poka Yoke	No	€720,000 per line, applied to every line ever installed

SMED	all zero	not purchasable before Year 2; it is the first Year-2 purchase

Raw material is production × 1.02, measured exactly in Year 1, with a further 5% on S only as cover for the Y2 shortfall. In test Y1 this field was keyed as 130,600,000 instead of 13,923,000 and cost €1,576,157 net and second place. **Two people check it, and the checker is not the typist.**

10 Finance

Field	Value	Why
Loan requested	27,000,000	capex is paid in cash on 2 January, leaving €3,000,000; the trough is –€21,080,961 at the end of February, before the first 60-day receipts
Line of credit	10,000,000	€100,000 for 1.8× cover of the trough, against a 20% overdraft
Fixed-term deposit / investments	0	there is no surplus until the December close; in test Y1 a €15,000,000 deposit earned nothing and cost a €150,000 penalty
Factoring	No	it would cost about €2,700,000; §41 adds receivables back to available cash on 2 January, so declining costs only intra-year timing
Customer terms	60 days	shortest option, smallest receivable
Supplier terms, S and H	120 days	the raw material bill is €41,664,578; four months of it is free financing through the trough

4 · What the plan produces

Projected income statement	€
Revenue at retail	333,136,287
less retailer margins	–36,837,998
NET REVENUE	**296,298,289**
less cost of goods sold	–94,359,202
GROSS MARGIN — 68.2%	**201,939,087**
advertising 678,000 · POP 48,000	–726,000
sales managers 600,000 · reps 2,592,000 fixed · 1,553,564 variable	–4,745,564
office payroll · general overhead	–3,418,000
market research · preventive maintenance	–981,800
logistics (trunk 211,200 · last mile 1,800,000 · platforms 1,100,000)	–3,111,200
research and development, 4.5% of retail	–14,991,133
depreciation, 12% of lines	–6,120,000
workforce turnover, at the test-round rate	–306,000
OPERATING INCOME	**167,539,390**
loan interest, fee and credit line	–3,070,000
NET INCOME — 55.5% of net revenue, ROE 71.4%	**164,469,390**

The §41 test	€
80% of new fixed assets 51,000,000	40,800,000
50% of media, POP, managers, rep fixed pay, overhead and research	2,499,900
15% of raw material 41,664,578	6,249,687
OCR required	**49,549,587**
Cash available 2 January	54,000,000
Headroom — the simulator will not borrow on our behalf	**+4,450,413**

This headroom is why the plant stops at 30 lines. 12 S + 20 H earns more in the central case but leaves only €780,000 of OCR headroom, and a single change to the Director's raw material price would push the plan into an automatic loan.

Robustness — six demand scenarios and a demand shock

Scenario	Net income	S sold	H sold	Unsold to inventory
B's sunscreen starved of media, H elastic at –1.30	132,683,215	4,680,855	24,314,489	6,104,656
B's sunscreen weak, H elastic at –1.30	149,312,542	5,850,000	24,314,489	4,935,511
Central — H at –0.90, S at –1.50	164,469,390	5,850,000	27,076,840	2,173,160
Sunscreen strong in B	163,296,126	5,850,000	27,076,840	2,173,160
H inelastic at –0.563	172,897,815	5,850,000	29,250,000	0
S very elastic at –2.50	160,117,029	5,850,000	27,076,840	2,173,160

H demand 30% below the central curve at every price	**113,919,529**	**—**	**—**	**—**

The worst of these still earns **27 times the test round's Year-1 result** and beats Plan B by €28,745,273. Unsold units are not a loss: they become finished-goods inventory, never deteriorate, and are sold first the following year.

Four-year path

Year	Plant	Utilisation	Produced	Sold	Revenue	Net income	Accumulated
1	10S+20H	81.25%	35,100,000	32,926,840	333,136,287	164,469,390	164,469,390
2	11S+21H	92.9%	42,473,880	33,799,943	349,033,971	173,101,883	337,571,273
3	11S+21H	92.9%	42,473,880	33,799,943	349,033,971	174,001,883	511,573,156
4	**11S+21H**	**92.9%**	**42,473,880**	**33,799,943**	**349,033,971**	**174,901,883**	**686,475,039**

Year 2 buys two SMED modules for €6,000,000 — the measured 92.9% utilisation lifts output 14.3% on lines already owned — and fills the last two line slots for €3,600,000, funded from the Year-1 close. The loan repays a third a year. Plan 2.1 projects €464,954,028 over the same four years on its own uncalibrated cost basis.

5 · Before submitting

- Open **Information** first and copy the Director's actual values. Per-capita income, raw material and power costs, exchange rate and every interest rate can differ from the test round. Loan maturity already fell from 4 years to 3 between the test years. If anything has moved, the plan is re-run before it is entered.
- Sales managers 6 / 6 / 0 — resets to zero every year.
- Logistics: 12 platforms in A and B; trunk 2 × 24,000 kg each; last mile 30 in A and 20 in B at 10,000 kg; market E at its form minimums, never zero.
- All six salary fields re-entered; training department No.
- Reps A 4+4, B 5+3, salary 27,000 and 0.60% in both markets.
- Every channel we sell in carries a price, a retailer margin **and** a shelf-space percentage. Market E blank everywhere except logistics.
- Campaign 18 for S and 19 for H, in **both** markets. Media identical in A and B.
- Lines to install 10 S and 20 H; **lines to be activated 10 and 20** — this field resets to zero.
- Shifts 3; preventive maintenance checked; Poka Yoke unchecked; SMED all zeros.
- **Raw material: S 6,265,350 and H 29,835,000. Checked by a second person.**
- Loan 27,000,000; credit line 10,000,000; deposits 0; factoring unchecked; customers 60 days; suppliers 120 days both products.
- OCR 49,549,587 is below the 54,000,000 of cash.

Open risks, stated plainly

Risk	Size	What bounds it
The Year-2 readings are second-hand and could not be verified from source files	the whole demand model	four independent consistency checks passed; and the break-even in section 2 holds without any elasticity
H at 8.00/8.75 is 3–13% above the highest price ever observed	up to €50M	market A's share would have to fall below 25% before 5.25 was better
S at 18.00 is 24% above the highest price observed	€16M	18.00 is the peak even under the most S-elastic scenario tested
Last-mile vehicle count is an estimate, not a rule	€1–2M	30 and 20 sit above the Y2 evidence floor; the Y1 result reports the cost and the flags
Sunscreen demand exceeds output by 873,103 units	§34 brand damage	unquantified; the alternative is 4 more S lines, which breaches the OCR test

Sources

day 1/year 1 DEP, REP, INV and INF result reports — every Year-1 figure, verified directly · day 2 - real thing/FINDINGS FROM TEST FOR THE REAL RUN.md — the calibrated constants and their derivation · 06-graded-year1-plan-2.1 and MMT39 - Graded Year 1 Plan 2.1 (8S-24H).pdf — the Year-2 readings, the trunk arithmetic and the last-mile evidence floor · YEAR 1 - OPTIMAL STRATEGY.txt — the HR, finance and research positions carried forward · Scenario MMT39 §3, §6, §10–§15, §17–§23, §25–§32, §34–§38, §41–§44, §48, §49 · 3 2026 MIM DD C2 EN MMT39.pdf — net rating and demand drivers · 4 2026 MIM evaluation guidelines.pdf — grading and the 18 September report deadline · model/mmt39.py plus the calibrated cost, capacity, demand and financing model built for this document. Standard cost, capacity, commission, R&D, depreciation and factoring each reproduce the simulator's own Year-1 figure exactly.